

Model Interpretation & Business Insights

Project: Mobile Money User Activity Classification — CSC 3221

A. Model Interpretation

The Random Forest classifier segments users into Low, Medium, and High activity groups with 78.9% average cross-validation accuracy. The model operates by learning decision rules from 19 behavioral and demographic features. Each of the 200 trees independently classifies a user, and the majority vote determines the final segment.

How predictions work (example): A user with `tx_per_month > 5`, `total_fees_paid > 3,000 XAF`, and `total_transactions > 60` would be routed to the 'High' segment by most trees. A user with `tx_per_month < 2` and `total_transactions < 15` would typically be classified as 'Low'.

Feature contributions: Transaction frequency (`tx_per_month`) dominates at 74% importance, followed by total fees (14%) and transaction count (8%). Notably, demographic features such as age, income, and education contribute less than 5% combined — behavior predicts segment better than demographics.

B. Business Recommendations

1. Personalized Loyalty Programs

High-activity users (≥ 6 tx/month) generate the most revenue through fees. MTN/Orange should offer them fee discounts, cashback rewards, or premium services to retain and grow this segment. Estimated impact: 10–15% reduction in churn for top users.

2. Onboarding Campaigns for Low-Activity Users

Low-activity users (avg 1.1 tx/month) use limited transaction types (avg 2.3 types vs 7+ for High). Targeted tutorials, transaction fee waivers for first 30 days, and referral bonuses could increase their monthly activity by 2x. This segment represents a 40%+ growth opportunity.

3. Business User Identification

Users with high send/receive ratios and business primary use should be offered merchant payment integrations, business accounts with higher limits, and micro-loan products. Transaction data can serve as a credit history proxy for the unbanked.

4. Rural Connectivity Investment

Rural users are disproportionately in the Low segment, partially due to connectivity challenges. Investing in USSD-based (non-smartphone) transaction features and agent banking networks in rural areas could significantly expand the active user base.

5. Weekend Engagement Campaigns

Weekend transaction ratios are uniform (28–32%) across segments, suggesting weekend promotions (e.g., zero-fee Saturdays) could differentially benefit all segments equally without segment bias.

C. Limitations and Ethics

Model Limitations

- Small sample size (60 users) limits generalizability. Model should be retrained on 500+ users before deployment.
- Self-reported transaction data introduces recall bias and approximation errors.
- The activity segments are derived from the same features used for prediction — circular definition risk.

- The model cannot account for temporary changes in behavior (e.g., seasonal spikes, COVID effects).
- No time-series modeling was applied; transaction trends over time are not captured.

Ethical Considerations

- **Privacy:** The model should never be used with personally identifiable data. All inference must be on anonymized user IDs.
- **Fairness:** Activity-based segmentation should not be used to deny services to Low-activity users. It should guide encouragement, not exclusion.
- **Transparency:** Users should be informed if their data is used to personalize services, with opt-out options.
- **Data minimization:** Only transaction metadata (counts, amounts, types) — not content or personal communications — should be used.
- **Potential misuse:** The model should NOT be used for law enforcement profiling, credit denial without appeal, or insurance pricing.

D. Future Work

- Collect larger dataset (500+ users) across all regions of Cameroon for better representation.
- Incorporate time-series features: trend in monthly activity, seasonal decomposition.
- Build a continuous credit-scoring model using transaction history as RFM (Recency, Frequency, Monetary) features.
- Develop a real-time prediction API using Flask/FastAPI for integration with financial systems.
- Partner with MTN/Orange Money for access to anonymized actual transaction logs.
- Apply cluster analysis (K-Means, DBSCAN) as an unsupervised complement to validate segment boundaries.